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Computer Law & Security Review

Artificial intelligence in the work process. A reflection on the proposed European Union regulations on artificial intelligence from an occupational health and safety perspective

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ARTICLE INFO

Keywords:
Artificial intelligence
Safety
Occupational health
Employer
Employee
European Union

ABSTRACT

Artificial intelligence (AI) finds increasingly growing applications in the working environment. Its importance has been recognised by the European Parliament and the European Commission, as reflected in the legislation prepared at the European Union level. As the use of AI creates new risks hitherto unknown from an Occupational Health and Safety (OHS) perspective, the question is whether the proposed EU regulations address these risks. The starting point for further consideration should be an analysis of the proposed changes to EU law in the context of the general principles of labour law. In addition to proposals to amend EU law on artificial intelligence, this article examines current occupational safety and health legislation. Issues related to occupational safety and health monitoring of employers using artificial intelligence were also the subject of the study. The social sciences' perception of human labour is not insignificant in assessing the new relationship at the employer-AI-employee level. The proposed model for regulating AI by the EU legislator is insufficient. First and foremost, there is no clear indication of the employer's obligations towards employees concerning occupational health and safety. Certainly, the essence of the EU law should be to establish the role of the employer in the process and of the Labour Protection Authorities, on the assumption that AI is only a working tool and not a subject of the law. Authorities should work together with employers to achieve the regulation objectives. Consideration should be given to introducing a responsive method of regulation in EU law, whereby the employer's application of employee health protection standards would be reviewed by the Authority to ensure that the objectives of the regulation are met. Depending on achieving a specific outcome or failing to meet the set targets, the Authority could intervene by applying appropriate punitive, incentive or reward mechanisms to the employer. Such a system of action, supported by providing adequate information to employers about the new risks associated with using AI, makes it possible to mobilise employers to take specific measures to protect employees' mental and physical health.

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https://doi.org/10.1016/j.clsr.2023.105825
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1. Introduction

Globally, despite the COVID-19 epidemic, the war in Ukraine and food problems, especially in African countries, there is a trend of increasing use of new technologies in everyday life. Indeed, it is hard to imagine, especially in Europe, not having a computer, the internet, access to email or computer software that calculates employees' salaries. These scientific and technological advances improve the functioning of society and enhance individual performance (Deranty and Corbin, 2022). On the other hand, in order to find one's way in the digital society, one must master the mechanisms of new technologies, which is a challenge, especially for the older generation.

Artificial intelligence is an example of the great importance of scientific and technological advances in a global society. But what is AI?

Artificial intelligence is defined in different ways (Čerka et al., 2017). In the literature, AI is defined as the ability of a computerised system to make decisions instead of humans (Tuon et al., 2022). Its operation is made possible by machine learning based on data processing (Grosz and Rolland, 1990). There are various types of machine learning. One is to acquire knowledge based on observation and actual interaction (Buiten, 2019). On the other hand, an AI system is a software-based system or part of a device that displays behaviour that simulates intelligence by, among other things, collecting and processing data and analysing and interpreting the environment. The system then takes autonomous decisions to a certain extent for a specific purpose (European Parliament, 2020a).

The development of AI brings with it many opportunities, but the serious risks arising from it should not be forgotten either. Artificial intelligence is having an increasing impact on employee health and safety. The European Union should ensure that appropriate regulations are in place to meet the new challenges posed by the development of AI in particular. The European Union has recently been taking specific action on this issue. It is, therefore, crucial to determine whether the European legislator's current legislative proposals are sufficient to protect employee health. Do they guarantee correct relations at the employer-employee level in the context of the right to safe and healthy working conditions and work ethic?

2. The significance of artificial intelligence in the work process

Artificial intelligence can be used in many ways in the work process. The literature indicates that the impact of AI on human labour comes down to the use of computational methods that rely on collecting and processing data to undertake specific actions. It can be applied, among other things, to coordinating machinery and industrial processes, managing the workforce (especially from an HR perspective), assessing customer risks and benefits, and analysing staff safety (Deranty and Corbin, 2022). Assessing employee performance using technological processes is one example of a new approach to work (Brown, 2017). Assessing an employee's potential future illness or the occurrence of an occupational disease after examining the employee's genes is also an example of the new goals for which AI is used (Flaherty, 2019).

Nowadays, we see the increasing importance of algorithmic management in labour relations, using AI mechanisms to plan or evaluate employees' work (Nowik, 2021). Artificial intelligence can fundamentally impact work management, even to the point where it can not only aid the work process but also manage it autonomously by making autonomous decisions based on the data collected. In such a situation, there is a new, hitherto unknown relationship between the employee and AI (Nowik, 2021).

Artificial intelligence can also be used directly to improve employee health. Wearing exoskeletons can improve workers' productivity and promote health protection. An exoskeleton is a robot that a human wears in the form of a suit, constructed with an appropriate structure that improves the dynamics of the user's limbs and joints (Sawicki et al., 2020). It allows the employer to conduct a biometric analysis of the worker and bring the worker to full fitness after injury (Ajunwa, 2019). The exoskeleton also relieves pressure on the spine. In South Korea, for example, such a device has been developed to support a person's feet, legs and lumbar region. It allows the worker to lift and carry heavier goods (LG's robot suit is straight-up sci-fi and will debut at IFA Berlin, 2018).

3. Challenges in the working environment related to the use of artificial intelligence

Despite the many opportunities to use AI in the work process to support employee efficiency, there are also some challenges related to OHS. While AI can positively impact the healthcare risk management process, there are risks associated with its use. The primary risk is still related to the potential occurrence of workplace accidents that result in an employee suffering an injury at work (Gaskins, 2004). Even exoskeletons equipped with AI to assist a worker in movement or rehabilitation can cause injury when malfunctioning. Hence, it is advisable to put in place an appropriate occupational safety and health policy when using them (Ajunwa, 2019). It is also not insignificant that robots equipped with AI may be used to provide labour, generating new, previously unknown risks in the working environment (Čerka et al., 2017). One aspect of their use in the work process can be the problem of these robots communicating properly with workers, which can pose a real danger of workplace accidents (Jarota, 2021).

Nowadays, the source of the accident is no longer just traditional mechanised manufacturing but, above all, the digital network advances that supervise the work process. A worker may suffer a physical injury due to the rush and time pressure resulting from the AI's work process management (Wigert and Agrawal, 2018).

The psychosocial risks associated with AI, which can affect the employee's mental health, should not be overlooked. These risks are evident above all in the employee's stress related to the provision of work, mainly when the nature of the work changes (Leka et al., 2015). Mental health is also affected by physical risks. Physical risks and psychosocial risks can intersect (Leka et al., 2015). For example, the lack of adequate rest during the day or week can cause mental illness

(Tandin et al., 2014). Research shows that even a few extra hours worked outside regular hours can increase the risk of an employee's health being compromised. The problem is particularly acute for remote employees working from home using electronic communications, where the boundary between work and personal life can become blurred (Arlinghaus and Nachreiner, 2014). Lack of rest can also result from working longer hours due to the fear of losing one's job because of new technologies (Deranty and Corbin, 2022).

Psychosocial risks in the workplace can arise from algorithmic management in the work process enabling employee control, (Parent-Rocheleau and Parker, 2022) particularly the potential awareness of employees of employers' access to their sensitive data. Theoretically, one could imagine a situation where genetic screening on the likelihood of future disease would be carried out before starting work and, therefore, before exposure to working environment factors. Based on this, the employer would refuse to employ the given individual (Field, 1983). Employers may try to use AI to collect information about an employee's health status to reduce the potential costs of sickness absence (Bernstein, 2006). The collection of an employee's data can, for example, be a source of discrimination against the employee (Ajunwa, 2019), and this can contribute to, among other things, depression (Gordon, 2021). The topical question remains what the employer will do with this knowledge in a particular case, whether it will ultimately end with the termination of the employment contract with an employee showing a higher propensity to illness, discriminating against them due to their health status. The threat of dismissal can cause additional stress to the employee. The fear of losing a job can negatively affect an employee's mental health and attitude. It can lead to conflict and professional burnout (Quilnan, 2007). On the other hand, information about individual employees makes it possible to anticipate adverse health effects in individual employees and prepare a plan for their treatment well in advance (Flaherty, 2009). Despite this advantage, knowing an employee's health characteristics must not cause the employer to abandon the primary objective of eliminating and reducing health risks to the employee in the workplace and must not constitute grounds for discriminatory action against the employee (Draper, 1996).

While employees may use medical information in making employment decisions (Draper, 1996), their data collection by employers should not violate their right to privacy (Levinson, 2010). While the legal protection of privacy is necessary, the collection of data and, based on it, the evaluation of employees, including their management, can lead to additional stress and competition between some employees. An employee could, even in the case of legitimate sickness, provide work for fear of termination. Consequently, this may prolong the illness's duration or even make it chronic (Quinlan, 2007). If this problem affected many workers in the same geographical area, it would impact the local healthcare system. Such a spiral of stress and prolonged illness is detrimental to employees, employers and the healthcare system. Therefore, these risks are not just a challenge at the employee-employer level. Given the need for public healthcare treatment of possible medical conditions, especially chronic ones, arising from work, these risks may affect the general public (Lambert, 2003; Gaskins, 2004). In Europe, this problem should be of particular note due to the high number of deaths caused by mental illness (Tandin et al., 2014).

4. Proposed legal solutions in the EU for artificial intelligence

The European Union notes new risks in the working environment that employers should address, for example, in the 'European Pact for Mental Health and Well-Being' (Slovenian Presidency of the EU, 2008) or 'The EU Strategic Framework on Health and Safety at Work 2021–2027. Occupational Safety and Health in a Changing World of Work' (European Commission, 2021b). When reviewing the current proposals for solutions to regulate AI, it should be noted that it has been under discussion in EU bodies for several years. In October 2020, the European Parliament adopted specific resolutions related to AI, two of which specifically address safety in using artificial intelligence. In its resolution of 20 October 2020 on a framework of ethical aspects of AI, robotics and related technologies (European Parliament, first resolution, 2020), the European Parliament called on the European Commission to, among other things, propose a comprehensive regulatory framework covering ethical principles and legal obligations related to the development, deployment and use of AI, robotics and related technologies in the EU, including software algorithms. The resolution stressed that new legislation should be based, in particular, on making AI, robotics and technology human-centric, human-made and human-controlled. In addition, according to the EP, potential regulations should aim to ensure that technologies are used safely, that users are informed that they are interacting with AI systems and that they can access the information they may need to operate them. The regulations should guarantee, in the event of non-compliance with the specified safeguards, a temporary deactivation restoring the safe functionality of the device. In another resolution of 20 October 2020 with recommendations to the Commission on a civil liability regime for artificial intelligence, the European Parliament (European Parliament, second resolution, 2020) advocated that, due to the significant potential for causing harm, all operators of high-risk AI systems listed in the annexe to the proposed regulation should have civil liability insurance. Furthermore, the EP pointed out that an AI system poses a serious risk when its autonomous operation involves a significant potential to cause harm to one person or a group of persons in a random manner and beyond what could reasonably be expected. Therefore, in the EP's view, if a user, i.e. a person who utilises an AI system, is a participant in a harmful event, he or she should only be liable under this Regulation if he or she can be considered an operator. However, the affected person should have the right to make liability claims throughout the liability chain and throughout the life cycle of the AI system. In general, according to the resolution mentioned above, liability refers to third-party liability cases where an AI system operates in a public space and, consequently, many persons are at risk. The EP assumes that in such a situation, the affected persons will often be unaware of the operation of the AI system and will not have any legal or contractual relationship with the operator. In the event of damage, affected persons will be able to

make fault-based liability claims against the AI system operator, so it will be their responsibility to prove the operator's fault (European Parliament, 2020b).

The European Union has recently taken some legislative action to regulate AI. This is reflected in the Proposal for a Regulation of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain union legislative acts (European Commission, Proposals, 2021c). It introduces new definitions; for example, an artificial intelligence system is software developed using one or more techniques and approaches listed in Annexe I (European Commission, Annexes, 2021a). This software may, for a given set of purposes defined by a human – generate outputs such as content, predictions, recommendations or decisions that affect the environments with which it interacts. Under the EC and EP proposals, a provider is a natural or legal person, public authority, agency or other entity that develops an AI system or has it developed with a view to marketing it or putting it into service under its own trade name or trademark – in return for payment or free of charge. In turn, a user is a natural or legal person, public authority, agency or other entity that uses an AI system under its control, except when the AI system is used as part of personal, non-professional activity. These proposed regulations should include employees and those working through online platforms, as referred to in the 2021 Commission work programme. As a general rule, such persons should not be considered users within the meaning of this Regulation (European Commission, Proposals, 2021c).

The proposal under examination takes into account, in principle, the recommendations of the European Parliament indicated above. Above all, the regulation aims to protect the rights guaranteed by the Charter of Fundamental Rights (Charter of fundamental rights of the European Union, 2012), including the right to human dignity, respect for private life and the protection of personal data, the right of workers to fair and just working conditions and the right to a high degree of health and safety protection. This is to be ensured, in particular, by a control mechanism. Artificial intelligence is to be controlled by humans. However, this only applies to high-risk AI systems. Indeed, according to Article 14 of the proposal for a regulation, high-risk artificial intelligence systems will be designed and developed in such a way, including by incorporating appropriate human-machine interface tools, that natural persons can effectively supervise them during the period of use of the high-risk AI system in order to prevent or minimise risks to health, safety or fundamental rights. Such systems include those used in the area of employment, employee management and access to self-employment, in particular for the recruitment and selection of candidates, for promotion and termination decisions, and for the assignment of tasks, monitoring or evaluation of persons in contractual employment relationships (European Commission, Proposals, 2021c).

The European Parliament and the European Commission see certain risks arising from using AI, including in the recruitment and assessment of employees, and the danger of discriminating against employees arises with its use. Artificial intelligence systems used to monitor performance and behaviour can also significantly impact employees' data protection and privacy rights. Therefore, the EU bodies propose, based on Article 15 (1) of the proposal, to introduce the principle that high-risk AI systems should be designed and developed in such a way that they achieve, given their intended use, an appropriate level of accuracy, robustness and cybersecurity. Furthermore, they should perform consistently in these respects throughout their life cycle.

Article 16 of the proposal for a regulation provides for the obligations of providers of high-risk AI systems, whereby they are to ensure that their high-risk artificial intelligence systems comply with the relevant requirements laid down in the Regulation, have a quality management system that complies with certain principles described in the Regulation, and have drawn up technical documentation of the AI system. The proposed EU law did not introduce such obligations explicitly for the employer; nevertheless, assuming that the user is the employer, it can be assumed that the employer could have identical obligations as the provider in certain situations. However, these would be the exceptional cases referred to in Article 17 of the proposal. Such tasks would only have to be carried out by the employer if it were to market or commission a high-risk AI system bearing a trade name or trademark, if it were to change the intended use of that system, or if it were to make significant changes to that system. In such a situation, the de facto responsibilities of the provider are transferred to the employer having, as it were, the status of the user and provider at the same time (Cefaliello and Kullmann, 2022). Under Article 29 (1) of the draft regulation, the employer is obliged to follow the accompanying instructions for use of AI systems as a high-risk user. In addition, under Article 29 (4) of the proposed regulation, employers should monitor the operation of that system based on the instructions for use. If they suspect that using that system following the instructions for use may lead to risks within the meaning of Article 65 (1), they should inform the provider or distributor and stop using the system. An analogous information obligation exists if a serious incident or system malfunction is identified. According to the EU proposal, the employer as a user should also keep records of events automatically generated by the system. Notwithstanding these obligations, the employer should ensure safe and healthy working conditions with artificial intelligence systems. Under Article 29 (2) of the draft regulation, the general obligations arising, among others, from the Council Directive 89/391/EEC of 12 June 1989 on the introduction of measures to encourage improvements in the safety and health of workers at work (Council Directive 89/391/EEC, 1989) remain in force. The fundamental obligation under Article 6 (1) of the directive above to take the measures necessary to ensure the safety and health of workers, including the prevention of occupational hazards, therefore also applies to responding to risks arising from the use of artificial intelligence in the work process.

The proposed law in the EU does not restrict the employer's use of OHS measures. It is crucial to achieve the stated goal of protecting the employees' health in the working environment, which can be challenging for employers from the perspective of the new risks associated with artificial intelligence. Furthermore, artificial intelligence should be supervised by the employer or those acting on the employer's behalf. The proposed regulations allow that to implement human surveillance of artificial intelligence indicated by the provider, the employer is free to use the means to achieve this goal. For example, the

employer could put in place procedures for the internal control of the AI system by ordinary employees and their supervisors to maintain oversight of the system as recommended by its provider. The employer could also introduce company dialogue on controlling artificial intelligence in the context of OHS risks.

The EU regulation proposal also introduces notification mechanisms for high-risk AI systems. Each Member State will be responsible for designating or establishing a notifying authority responsible for developing and applying the procedures necessary for the assessment, designation and notification of bodies that assess and monitor high-risk AI systems in the context of legislation.

5. Artificial intelligence issues from a labour law and ethics perspective requiring in-depth reflection by the EU

To analyse the problems of regulating AI in the working environment, it is necessary to explain the essence of labour law. Undoubtedly, labour law is based on the relationship between humans: the employee and the employer, the guarantee of the employee's dignity by ensuring him, among other things, safe and healthy working conditions. This follows from documents adopted by the European Union and the International Labour Organisation. For example, the very first ILO document, i.e. the 1919 Constitution (ILO, Constitution), included as one of the main reasons for the introduction of the Constitution the need to protect workers in case of illness, occupational diseases and accidents caused by work. The Declaration of Philadelphia, adopted in 1944, introduced the principle that labour is not a commodity in its first point (ILO, Declaration). In turn, under Article 31 (1) of the Charter of Fundamental Rights of the European Union every worker has the right to working conditions that respect his or her health, safety and dignity.

Human dignity and human security must be looked at not only from the perspective of the law but first and foremost with a human work ethic in mind. Human dignity is the value that distinguishes human beings from any other creature (John Paul II, 1981; Cebulla et al., 2022). The social sciences point out that the worker's dignity should be the starting point for legal considerations of OHS in the context of new technologies, and to protect it, the view of the employer-employee relationship should be based on the human aspect. It means that the basic assumption should be that the employee-employer relationship is a human-to-human relationship. According to Catholic social teaching, the organisation of work itself should consider, among other things, the well-being of the employee so that the employee has the appropriate conditions to carry out daily tasks. In turn, the equipment should meet the demands of production and human needs. The work process is done by the people, and its outcome is for the people. It is important to remember that, apart from the emotionless machine, it is primarily the human being who works in the workplace (Wyszyński, 2021).

The International Labour Organisation, seeing the risks posed by, among others, AI, calls on states to regulate labour law in a spirit of dialogue with trade unions and employers' organisations, taking into account the new scientific and technological challenges. In its report entitled "Work for a brighter future", the ILO makes it clear that in the future, people should continue to be central to the work process, and one element of the implementation of the strategy should be the recognition of occupational safety and health as a fundamental principle and right at work (ILO, 2019). In this context, the question that would need to be answered is whether an AI system can be held responsible for the state of OHS in relation to the damage the system will cause.

For such responsibility to be attributed to AI, it would have to be granted legal personality. Indeed, the acquisition of legal personality generates by law the rights and obligations of the entity that holds it. Such an entity can thus enter into legal relations with other entities even if it is not a natural person (Solaiman, 2017). The main argument for endowing artificial intelligence with a legal personality is the increasing number of human-specific characteristics that AI possesses (Chesterman, 2020). According to this concept, it may be increasingly difficult to establish that the artificial intelligence's actions and decisions are performed on behalf of the employer as a legal person and with its full knowledge. Indeed, these actions can be unpredictable for the employer (Hildebrandt, 2020). Forecasts suggest that AI may surpass human skills and achieve self-awareness in the future (Chesterman, 2020). Due to the lack of adequate knowledge among employers, it would seem that software developers, owners or device manufacturers could be held responsible for artificial intelligence. However, establishing liability for damage may also be difficult on the part of the system developer, software developers and owners due to the autonomy features of an artificial intelligence system, which may act in an unplanned manner. Artificial intelligence can learn from experience and solve difficult, complex problems differently than its creators intended, even though the source code of its programming does not change (Calo, 2016). The main challenge is establishing a causal link between the damage that has occurred and the act or omission of those persons. Such a link may be difficult to prove in some cases, especially since artificial intelligence may be designed by multiple entities (Banteka, 2021). For this reason, the literature advocates giving artificial intelligence the status of a legal person so that it can be held responsible for its own actions (Asaro, 2012). Since legal personality has been granted to entities that are not natural persons, all the more so can a robot equipped with artificial intelligence have such status (Asaro, 2012). The rationale for this position includes the argument that artificial intelligence can commit crimes on its own and should therefore be accountable for them. According to this conception, there is no difference between a company with legal personality and a robot equipped with artificial intelligence (Hallevy, 2013; Asaro, 2012). Such a view is opposed by the claim that de facto giving legal personality to artificial intelligence will not absolve humans from responsibility for it. The economic costs of the associated damage will continue to be borne by the individual, who will have to purchase appropriate insurance (Solaiman, 2017). This is because granting subjective rights to artificial intelligence raises the problem of how to enforce the fulfilment of duties against it and who would represent it in a dispute with an employee before a court (Bryson et al., 2017). In addition, a typical legal person, e.g. a company, is made up of humans as opposed

to an AI-equipped robot, which is just a machine (Solaiman, 2017).

On the other hand, AI can draw conclusions based on data. Still, without understanding the meaning of the data, the values or feelings typical of humans are alien to it (Hárs, 2022). Human labour is an essential social element, and the workplace is a place to meet other human beings (Cebulla et al., 2022). Scientific and technological progress helps the worker, but it can lead to rivalry between workers, which can affect social peace. Labour protection is not only of social importance but also of economic or even systemic significance (Wyszyński, 2021). If humans are eliminated from the work process and replaced entirely by artificial intelligence, work may be treated as impersonal (Donati, 2020). For example, working exclusively with an AI-equipped robot superior to a human being in terms of intelligence can lower an employee's self-esteem. The employee may attempt to compete with artificial intelligence systems; in turn, these systems may affect the OHS of the employee. They can assess the employee, plan their work and make mistakes in doing so. Artificial intelligence is not a human person but merely a human tool. Work tools are not liable for damages caused in connection with their use. From this perspective, it is difficult to assume that it is not necessary to regulate employee health and safety explicitly because AI will be responsible for its actions and decisions.

The above assumption is correct from the perspective of the human work ethic, according to which the worker comes first before matter and thus products of new technologies. Unlike other creatures, only human beings are called to work (John Paul II, 1981). Thus, it can be said that the outcome of a human being's work is an AI system, while a human being is not a product of AI. Taking the opposite view reduces the worker to an element of efficiency and economy, while the personal sphere of the human being is lost (Duda and Świerczek, 2014).

The personal dimension of work is particularly emphasised in the philosophical sciences. Karol Wojtyła, later Pope John Paul II, pointed out that the person should be linked to the act as a conscious human action. According to this concept, the human being acts as a self. A person not only acts consciously but, more broadly, is aware of his or her decisions and knows who performs given actions (Wojtyła, 1969). Thus, in answering the question of whether a person must be given a status in law to be a person, representatives of this strand of philosophy point out that a person is the raison d'être of law. On the other hand, the law is not the basis of human existence. A person can only assert his or her rights against another person. The law is a relationship between persons committed to the common good (Krąpiec, 1986).

In this context, it is important to note another problem in protecting employee health. The literature additionally emphasises that employers may compensate for more serious OHS risks with appropriate compensation, e.g. bonuses (Lambert, 2003). This would consist of individual employers failing to provide information on preventive measures and applying such measures but compensating workers in return for failing to do so, shifting the OHS risk to them. This approach can even reduce the employer's costs when OHS expenses are higher than the bonus payment (Lambert, 2003). This is all the more convenient as the number of risk factors in modern working environments is increasing, posing a particular challenge for employers (Gaskins, 2004). However, this concept shifts additional responsibilities to the employee, who often has less organisational, technical and asset capacity to respond adequately to risks or know about them at all.

Shifting the responsibility for OHS from the employer to the employee or AI is incompatible with the general concept of worker health protection, according to which the employer is responsible for the health and safety of the working environment (Liu, 2020). The rule of thumb is that OHS is the responsibility of those who control the risks and can prevent or reduce them. The employer is such an entity (Bluff and Gunningham, 2003). This is not the employer's liability independent of fault ('*in the United Kingdom case*') (Case C-127/05 Commission of the European Communities v United Kingdom of Great Britain and Northern Ireland 2007). A rational approach should provide that, in the first instance, the employer is responsible for the damage caused to the employee by the AI and for the health and safety conditions while using it. The employer can only avoid this responsibility under Article 5 (4) of Directive 89/391/EEC, if the incident in question is due to extraordinary and unforeseeable circumstances beyond the employer's control or to exceptional causes which could not have been avoided despite the exercise of due diligence. However, the employer should demonstrate that any of the above is the case. According to the above directive, the employer should react to specific incidents, adopt measures appropriate to the circumstances and preventive actions and adopt a coherent prevention policy. Accordingly, the employer should reduce, according to the knowledge obtained, among others, from the AI system provider, the risks associated with the use of AI in the work process, adapting the working conditions of the employees to the circumstances. As a general rule, worker protection measures should correspond to the level of risk in the working environment for which the employer is responsible. Indeed, it is the employer's task to properly assess and control the risks (Colosio et al., 2017).

The European Commission's proposal for regulation on AI does not directly address the employer-employee relationship at all, which may blur the issue of responsibility for safety and health in the workplace. The principle is the supplier is liable, and the employer is only required to use the AI in accordance with its instructions for use. In the proposed EU regulation, it appears unreasonable to point out that only the provider is responsible for AI in principle, despite its indisputably important role in making the device safe. In order to send a clear message regarding AI and the associated risks to which the employer should respond, the employer should also be explicitly identified as being responsible for safety and health conditions in a working environment where artificial intelligence is employed. The general principle of the employer being held financially liable for, among other things, accidents at work should be the starting point for further consideration of the role of AI in the work process. The obligation of employers to regulate compensation for occupational accidents, in particular, motivates them to take all possible preventive measures, including looking for ways to solve OHS issues (Lambert, 2003).

The proposed regulations introduce a similar health protection model to the so-called Machinery Directive (Directive 2006/42/EC). The directive provides for several marketing the product obligations but does not directly address the employer's obligations towards the employee. Indeed, according to this directive, the manufacturer, like the AI provider, before placing the device on the market or putting it into service, ensures, among others, that it complies with the relevant fundamental health and safety requirements set out in the corresponding annexe or provides, in particular, the necessary information, such as the instructions for use. However, in line with the objectives of the directive set out in (12), the putting into service of machinery within the meaning of the directive can relate only to the use of the machinery itself for its intended purpose or for a purpose which can reasonably be foreseen. With the development of artificial intelligence, a fundamental problem is the lack of complete predictability of the effects of artificial intelligence systems on devices. Autonomous, automated artificial intelligence operations make it possible, for example, to monitor employees and impose sanctions on them for their inefficiency (Adams-Prassl, 2019). Even if a given machine equipped with an AI system were checked against basic OHS rules, there is a risk that it would cause new phenomena that did not exist before. It is difficult to predict all the implications and risks of artificial intelligence because of its autonomous operation, which can change. Therefore, the decisions of an artificial intelligence system can affect employees and be beyond their control. A worker can talk and communicate with artificial intelligence to perform work. The artificial intelligence system becomes even more accurate through machine learning based on processing data from conversations and collaboration with humans. Nowadays, artificial intelligence can also perform more tasks whose quality can also be improved. An ordinary machine does not function this way. More and more tasks performed by artificial intelligence are leading to new interactions between the employee and the system. These, in turn, affect the employee's working conditions in terms of both physical risks and psychosocial challenges related to, among other things, the employee's fear of losing their job. Equipping a machine with artificial intelligence can undoubtedly change an employee's working conditions. Therefore, a key challenge may be developing clear and safe rules for human collaboration with the AI system.

EU law provides for general OHS obligations. Under Article 6 (1) of Directive 89/391/EEC, the employer should respond to needs and adapt measures accordingly, taking into account changing circumstances and preventive measures to improve the existing situation. The employer should fulfil this obligation, taking into account the principles set out in Article 6 (2) of the directive. One of the rules is to take preventive measures in such a way that they are adapted to technological advances. According to the regulation in question, the employer should also provide employees with appropriate instructions or develop a coherent prevention policy that considers technical issues, ways of organising work, working conditions, social aspects and the impact of factors relating to the working environment. One of the employer's obligations under Article 6 (2) (d) of the directive is to select work equipment and working and production methods when carrying out work and reduce their adverse health effects. Directive 89/391 EEC thereby guarantees workers general mechanisms for protecting their health in the working environment, taking into account technological advances, including the development of artificial intelligence.

EU legislation extensively regulates employers' obligations to prevent risks and to provide the necessary organisational framework and measures to protect workers' health. Directive 89/391/EEC introduces general tools employers should use in response to OHS risks. They can also be applied in connection with the use of AI in the work process. Nevertheless, these are general measures. Meanwhile, working conditions have significantly changed since the directive mentioned above came into force, and the current challenges of using artificial intelligence are becoming more serious. This is somewhat related to the development of artificial intelligence (Jarota, 2021). The literature points to a classic example of artificial intelligence assigning tasks to employees and the time to complete them. Such a system can be programmed to meet the employer's needs, which does not relieve the employer of controlling how the algorithm works, if only in terms of work rate or safety. On the contrary, when an algorithm allocates tasks to employees disproportionately to their working time, over which the employees have no control, the employer should be able to modify the artificial intelligence system as appropriate to the circumstances (Cefaliello and Kullmann, 2022).

Artificial intelligence can make mistakes in performing its tasks. It represents a significant challenge to the protection of workers' physical and psychosocial health. Thus, following the equipment manufacturer's instructions for use or implementing general OHS procedures is no longer enough. Comprehensive and detailed solutions are needed to address the physical and psychosocial risks arising from the use of artificial intelligence. Given the dynamics of artificial intelligence development, which is not guided by feelings but makes decisions based on a computerised system, serious consideration should be given to introducing new OHS regulations in the European Union. The legislator should explicitly state in EU law the main risks associated with using artificial intelligence by employers and the recommended responses to them. Such a measure would allow for increased awareness among employers of fundamental OHS risks in the context of using artificial intelligence.

The proposed regulations also do not introduce obligations for employers as users in relation to workers to communicate OHS risks associated with using high-risk AI in the working environment. Admittedly, the general obligation to provide information on safety and health risks derives from Article 10 of Directive 89/391/EEC. Under Article 10 of the directive the employer has a duty to inform workers about OHS risks. This regulation does not relieve the employer of the obligation to communicate new risks to employees. Therefore, the directive provides tools to strengthen the process of informing workers about OHS risks related to artificial intelligence. It would seem that OHS risk communication to workers should not be limited to indicating the OHS risk itself but should also address how to respond to it. Employees should be aware of how the artificial intelligence system operates in the workplace. And even though the aforementioned directive introduces the

analysed obligation as already noted – the directive itself does not explicitly address the risks associated with, for example, using AI-based algorithms in the work process. Nor does the directive indicate what methods employers should use to ensure an effective flow of information between individual employees, especially between the management and ordinary employees. In large organisations, communication between such employees can be difficult. The EU regulations also do not put in place mechanisms to monitor employers' performance in communicating OHS risks to workers. Meanwhile, the information deficit among workers regarding OHS risks is fundamental from the perspective of protecting their health. This is because it should be kept in mind that there is an asymmetry of information about AI between rank-and-file employees and managers (Deranty and Corbin, 2022). However, a lack of complete information about risks also exists among employers. In turn, a systematic lack of information may result in inadequate measures being taken to protect health in the working environment (Lambert, 2003), which may justify intervention by the legislator through the harmonisation of laws at the EU level, to respond to new risks in a coordinated manner. In fact, the employer should communicate to the workers all the information obtained from the provider on AI affecting OHS, particularly all the associated risks, so that the workers are fully aware of them.

Workers' understanding of how AI works is crucial from the perspective of knowing the implications of an employer's use of AI and addressing the OHS risks associated with it. Adequate knowledge of its nature allows employees to adapt to new working conditions. Requiring employers to undertake individual AI awareness measures should be a fundamental starting point for changes to EU law. In practice, it appears that the general information obligation under Article 10 of Directive 89/391/EEC may not be sufficient. Specific mechanisms should be implemented to guide employers in fulfilling the above obligations to achieve the intended information goal.

Despite the extensive obligations under Directive 89/391/EEC, the practice to date of implementing EU OHS legislation may indicate that employers are not adequately monitoring the effectiveness of the measures in place to protect workers' health. There is a lack of a coordinated approach by European employers to OHS issues (European Commission, 2004). The regulatory practice of individual EU countries also varies (Walters et al., 2022). Some, for example, highlight psychosocial risks, although these risks are not explicitly specified in EU legislation. Others do not address them (Leka et al., 2011). Moreover, in the case of new threats and psychosocial risks, there has so far been a lack of appropriate activity in their prevention (European Commission, 2004). Despite the OHS regulation in the EU, it has not been possible in Europe to develop a consistent model across all EU countries for significantly reducing work-related illness and accidents (Nicholson, 2002; Liu, 2019). The introduction of EU regulations dedicated explicitly to occupational safety and health concerning artificial intelligence is therefore also supported by problems in the consistent application by EU countries of the current measures under Directive 89/391/EEC. In terms of the employer's OHS obligations, the EU legislator should draw lessons from both the aforementioned directive and Directive 2006/42/EC and relate them to the new challenges of artificial intelligence. Whether the changes advocated above would consist of an amendment to the draft regulation or the introduction of a separate directive dedicated to occupational safety in relation to artificial intelligence is an open and further question.

However, it is not enough to impose strict obligations on employers. Even if employers want to take appropriate precautions, they may not know about them. In addition, these measures may cost more than the potential gain to the employer (Lambert, 2003), which does not mean that they should be abandoned. The direct costs of accidents at work in the European Union include, in particular, the absence of workers, occupational disabilities, the need to pay survivor's pensions, medical expenses or the training of replacements (Ndjoulou et al., 2015). Ultimately, it may turn out in a particular case that, by neglecting OHS obligations, the employer will have to pay substantial damages many times the profit of the period in question. It would therefore be appropriate to regularly inform employers about the risks of AI and measures to mitigate risks in the working environment. This could be done by an appropriate agency, similar to OSHA in the United States of America (McGarity and Shapiro, 1996), and employers would then communicate OHS risks to employees. However, even such measures may not be enough. Providing only information without any legal mechanisms, incentives or motivation for employers may result in a lack of risk analysis due to, among other things, fatigue or the voluminous nature of the material to be consulted (Lambert, 2003). Therefore, it is essential to consider a broader view of this problem from the perspective of methods of regulating AI in the work process and the need to handle new OHS risks.

The dangers of AI in the work process require a new approach by employers and legislators. Above all, the way in which artificial intelligence functions in devices that can lead to an employee's injury requires particular consideration. The rivalry between employees, professional burnout due to employee evaluations and fear of losing a job due to an algorithmic assessment require adopting an appropriate operating model to eliminate or reduce risks. The problem is to ensure that the employee is guaranteed optimal working conditions with AI so that the pressure felt by the employee resulting from this collaboration is reduced and that the employee enjoys the guaranteed right to rest and collaborate with the AI system. Lack of rest or rushing can affect a worker's mental and physical health. Undue haste due to the organisation of work by the algorithm should be limited by appropriate procedures so that needless work injuries do not occur. If employees had more information about how the algorithm works (including the mechanism by which it collects and uses data), they might not find the new situation stressful. The fact that it will always be a human who oversees the autonomous decisions of AI would also be reassuring for the employee because, unlike AI, the employer can verify the other work indicators that the algorithm does not rely on. A human can also check if there is an error in the algorithm, so it is reasonable for a human to verify the final performance of AI.

The occurrence of potential accidents and illnesses that may result from using AI in the working environment must be reduced. This reduction can be achieved through the cre-

ation of law and setting clear rules for the employee on using AI in the work process.

The new OHS risks associated with using AI are societal, not just individual (Draper, 1996); therefore, the right approach to employing AI is also crucial from the healthcare system and social security perspective. Adopting appropriate mechanisms to protect the employee when working with AI is a specific task for the public authority in this context.

6. Methods of regulating artificial intelligence in the working environment from an OHS perspective

How to regulate AI in the work process? What OHS regulating methods should be adopted in this area?

In order to determine how OHS is regulated in relation to the risks arising from the use of artificial intelligence, it is crucial to clarify the possible role of external bodies with respect to employers' OHS practice. Indeed, the literature criticises the model of self-regulation of this sphere by employers without external control. Instead, it is pointed out that the approach to artificial intelligence should involve the active participation of an independent regulatory body, which should be given appropriate powers to monitor and enforce OHS obligations on employers (Yeung et al., 2019). And while there are various forms of this so-called self-regulation, whether giving complete freedom to the employer to regulate OHS without any control or with some internal or external control, the complete absence of external monitoring of employers' OHS practice may not be effective from the perspective of worker's health protection (Gunningham, 2011). This is also confirmed by the OECD, indicating that self-regulation, in general, may not achieve its objective due to, inter alia, a conflict between the private interest of the entrepreneur and the public interest or too much dependence of the internal control bodies on the audited entity (OECD, 2015). A specific intermediate solution combining self-regulation and control mechanisms proposed in the literature is to adopt the concept of co-regulation, whereby the employer introduces its regulations, but with the possibility of establishing them with the authority if necessary. This concept implies the simultaneous introduction of a power to inspect employers by an external body on the level of implementation of OHS obligations (Gunningham, 2011).

From the perspective of the many challenges posed by the new OHS risks of artificial intelligence, the concept of no oversight of employers' OHS activities cannot be accepted. Given the autonomous operation of artificial intelligence and its changing circumstances, it is impossible to leave the regulation of this sphere to employers outside of any external control. The decisions of the artificial intelligence system regarding employees may be beyond their control, which already requires the adoption of specific rules to monitor the process of applying artificial intelligence from an employee health perspective to prevent adverse developments for the employees' health. The new interactions between workers and the AI system require special activity. There seems to be a need to support employers through the involvement of expert bodies both in terms of recommendations on artificial intelligence and monitoring the fulfilment of OHS obligations by employers. The adoption of methods to control employers' OHS performance when introducing internal procedures remains an open question.

The first fundamental method of controlling OHS for employers using AI may be sanction-repressive method, which involve imposing obligations on employers on pain of sanctions for OHS violations. In general, according to this method, only penalties for law violations are applied to the employer. The premise of this concept is that only the application of severe sanctions makes it possible to fully ensure adequate working conditions (Schofield et al., 2014; Dorrian and Purse, 2011). According to views supporting the concept of regulating OHS in a repressive-sanctioning manner, the main argument in favour of its use is that it encourages the implementation of OHS measures on the part of employers to protect worker health (Ndjoulou et al., 2015). Nevertheless, there is then a lack of willingness on the part of the addressees to comply with the standards voluntarily. Moreover, as more regulations are introduced, the opportunities to find loopholes in the law may increase (Parker and Braithwaite, 2005).

Another approach is a method of responsive regulation, which seeks to find a less repressive measure than merely the typical sanction for non-compliance with the law. This method is not universal and depends on many variables; it takes into account the socio-economic context and the specific characteristics of the employer concerned (Ndjoulou et al., 2015). It presupposes that, in relation to the addressee of the legal provision, depending on the achievement of the intended objective, the Authority intervenes more or less according to the level of application of the law. The less frequent the targets are met, the greater the intervention; thus, there is a gradation in the Authority's activities (Ayres and Braithwaite, 1992). This intervention may also be referred to as enforcement. The Authority's activity is broader in scope and amounts to all activities designed to promote compliance of behaviour with the law and to achieve the stated objectives of the law (OCED, 2014). The authority monitoring the employer's fulfilment of its objectives should first seek to use persuasion and negotiation techniques to point the employer in the right direction. Sanctions should be used with far-reaching caution as a last resort. Therefore, as part of enforcement, the authority first encourages the performance of an obligation by soft measures. If these fail, it uses harsher methods, such as a ban on particular conduct. When these measures also fail to have a positive effect, monetary and non-monetary sanctions are used (Johnstone and King, 2008). One form of implementation of this method is the above-mentioned co-regulation, whereby the employer is obliged to implement its own regulations, but the state supports the employer in doing so and only interferes in the employer's activities if necessary (OECD, 2018).

The introduction of responsive regulation is promoted by the OECD, which also identifies measures that would strengthen the effect of this regulation method. Above all, enforcement must take into account the profile and conduct of employers. According to the OECD, the Authority's interventions should be synchronised and not duplicated. In addition, the process of interference by the Authority should be transparent, and the regulations that oblige a specific action should be clear to entrepreneurs. The whole system should

make it possible to verify the effectiveness of the application of the law from the perspective of the intended purpose and the stakeholders' perception of the regulation. The OECD proposes to introduce a system of ratings for entrepreneurs ranging from negative to intermediate to positive, depending on the severity of the problem (OECD, 2018). These assessments can be carried out, among other things, based on indicators of the law's effectiveness. They can also result in a specific form of incentives. In this way, in addition to sanctions for unsatisfactory performance, the employer could receive a reward in case of good performance. Indeed, an incentive can positively influence the employer's behaviour by encouraging the employer to implement appropriate safety measures (Bianchini et al., 2017). This incentive can be in the form of a certificate or a label communicated to the market, which positively impacts the reputation of employers and can motivate them to become more active in OHS (Montero et al., 2009).

Some challenges in applying this method may be the lack of regulation regarding the risks associated with artificial intelligence. Research indicates that national labour inspection authorities may have problems applying responsive regulation measures to risks the law does not explicitly address, such as psychosocial risks. This may be due to the lack of clear regulations and requirements for employers regarding these risks, which makes it challenging to demonstrate wrongdoing on the part of employers. It is difficult to use soft measures, such as persuasion, appropriately in such circumstances, the use of which requires adequate knowledge and involvement of the authority (Johnstone et al., 2011).

To conclude this discussion stage, the starting point of the responsive regulation approach is that the classic deterrence mechanism of merely issuing orders under threat of punishment through the common law is ineffective (Parker and Braithwaite, 2005). Despite the above position, doubts arise about using this approach for new OHS risks. However, these concerns should not disqualify this approach, which also offers the possibility to use harsh sanctions and not only soft measures. Lack of knowledge about new OHS risks seems to affect soft and hard measures similarly. Using one or the other methods of influencing the employer requires adequate knowledge and involvement of the authority. However, the more choices for influencing the employer, the better. The broader range of tools used, such as persuasion, punishment and reward, among others, makes it possible to adapt to the circumstances. The problem in applying this approach is not the measures themselves, which can be used, but their selection, which is influenced by the complexity of artificial intelligence functioning, which, in practice, may limit the authority's response.

This method requires more reflection on its potential for practical application. The legal protection of employee health in the context of artificial intelligence development should be addressed comprehensively. This means that, with appropriate enforcement methods in place, comprehensive action needs to be taken on the systemic regulation of occupational safety in relation to employers' use of artificial intelligence. Committing solely to the responsive regulation approach, without significant changes to EU law, could fail to have a positive effect on workers' health protection. Therefore, new risks should be addressed by the EU legislator, taking into account the specificities of artificial intelligence. EU law should emphasise the OHS risks associated with using artificial intelligence by employers to respond to them appropriately.

It is worth considering a responsive approach to OHS, bearing in mind the specifics of AI and its risks. A specific general action model could be adopted in EU legislation, delegating clearly identified tasks to employers. The Authority could recommend specific patterns of behaviour in OHS, taking into account the aspect of using AI in the work process. Employers, in turn, could introduce their own safety and health policies, taking into account the Authority's objectives and guidelines. These policies could be presented to the Authority for approval, which would reinforce, in this complex matter, the substantive assessment of the validity of the solutions adopted in OHS procedures. Then, based on reports and records of events, certain measures could be applied to the employer in connection with the application or non-application of the adopted rules. For example, the employer would be positively assessed, which would entail the possibility of using AI in the working environment, or would be given binding recommendations to comply immediately with the adopted pattern to protect the employee's health or, in the case of significant infringements, even be subject to specific penalties. This kind of cooperation between the Authority and the employer built through dialogue would make it possible to adapt the law to changing circumstances resulting, among other things, from scientific and technological progress, all of the effects of which are still difficult to foresee. Naturally, the very mechanism of interaction between the Authority and the employer would require in-depth reflection, taking into account the OHS challenges, the objectives of the regulation and the methods of monitoring the effectiveness of employers' compliance with the adopted model of action.

For the creation of suitable working conditions, it is desirable to involve employees, employers, authorities and the provider of the AI system. Their participation allows for a better understanding of the mechanisms of AI, the detection of AI risks and the involvement in the development of AI regulations (Mantalero, 2019). The provider should develop the system software, taking into account the risks in its use by employers. It should take into account, in particular, the worker's real work pace, the difficulty of communicating with the robot, the human influence on the operation of the artificial intelligence with the possibility of modifying it, and the potential of their control by an external body. Employers, on the other hand, should verify the operation of the AI against OHS rules and, if necessary, change its operation to fulfil the obligations under Directive 89/391/EEC correctly. Employees should be fully informed as to the risks associated with artificial intelligence. In turn, employers should listen to employees' concerns about artificial intelligence operations.

It is not insignificant that the general principle of Directive 89/391/EEC is that the planning and implementation of new technologies should be subject to consultation with the workers or their representatives where this may have a consequential effect on the health and safety of workers and concerns the choice of equipment, work procedures and determines the specifics of the workplace. Article 6 (3)

(c) of Directive 89/391/EEC imposes a specific obligation on the employer to guarantee the right of workers to consult the employer on these OHS issues. In order to ensure that the way in which artificial intelligence works is transparent, the participation of trade union representatives in the regulation of employer OHS concerning artificial intelligence is warranted (Cefaliello and Kullmann, 2022). Employers should seriously consider the need for a dialogue on AI in the work process with trade unions as a representation of workers, which can positively influence, among other things, the exchange of information on new risks in the work environment and how to respond to them. Studies show that the consequences of accidents at work are less serious, and OHS conditions are safer when trade unions actively participate in the OHS management system (Colosio et al., 2017; Menéndez et al., 2020). Involving employees in the work management process positively influences their attitude towards work. It gives them the feeling that still, despite the autonomy of artificial intelligence, it is the employees who have a tangible impact on the work organisation (Lee et al., 2021).

The company dialogue allows for creating a space for consultations to protect workers' health, including the introduction of internal OHS regulations at employers that are acceptable to the workers. The premise should be that employers put in place appropriate internal regulations on using artificial intelligence in the OHS context in consultation with trade unions to comply with legal obligations. Subsequently, these regulations could also be scrutinised from the perspective of achieving their intended purpose by an external body within the framework of responsive regulation.

7. Conclusions

The introduction of the principle in the proposal of a regulation of the European Parliament and the European Commission that humans control AI and not the other way around is certainly welcome. Artificial intelligence should not be a tool used beyond human control.

In the workplace, it is possible to note the occurrence of fears of being assessed by an algorithm or losing one's job due to the algorithm. Due to these concerns, employees may be more prone to conflict with each other than before. An AI system can also cause discrimination in the working environment. The use of AI in the work process can cause create communication problems between AI and humans. These risks can result in mental disorders and negatively affect the physical health of workers. However, European legislation and proposals to amend EU law have not explicitly introduced a model for protecting workers under the stress of working in new conditions involving AI, in context with the current threats. In addition, there are also hitherto unknown risks of mechanical injury in the workplace associated with the functioning of AI, exoskeletons being an example. Therefore, what is missing from the EU's side is not only a reference to psychosocial risks but also to the physical risks associated with AI mechanisms applied in devices. These new risks require an appropriate response from the European legislator in terms of worker health protection.

EU regulations should guarantee workers' right to safe and healthy working conditions that correspond to the employer's obligations. Artificial intelligence itself should be clear to everyone affected. Employees should have more comprehensive information on how AI works and the risks involved. Notably absent from the EU's latest AI proposal is a direct indication of a worker's right to safe and healthy working conditions when using AI in the work process. The current proposals regulation EU relating in principle to the obligations of the AI system provider do not correspond with the risks in the working environment and the health protection needs of the worker. In light of the above, it is legitimate to conclude that the model proposed by the EU for protection against the harmful effects of using AI in the proposal for a regulation is insufficient from the labour law perspective.

Thus, in answering the question of how to regulate AI in the work process, it must be recognised that the standard sanction-repressive method of punishing the addressees of legal provisions for violations does not necessarily work. Addressees of the regulations, against whom only sanctions for infringements are introduced, may deliberately avoid complying with their obligations under the law. New OHS risks, hitherto unknown, require special interaction between Public Authorities and employers. The answer to the development of a cooperative model is a responsive regulation method so that it is possible to verify the effectiveness of the employer's application of OHS standards from the perspective of their objective. It allows the law to be better adapted to the circumstances with a view not only to punishing violations but also to rewarding action in line with accepted patterns of behaviour. It would therefore be necessary to oblige the employer to implement internal regulations that should meet the stated objectives, subject to review. In this context, it is worth considering the introduction of broad powers of the Authority in relation to the employer. This body could reward or punish employers in certain situations, and make recommendations to them. Such an OHS-dedicated Authority could also provide information on the risks associated with AI and methods to reduce the occurrence of these risks. The implementation of the analysed approach depends on the specific risks in the working environment. This method is not universal and should be adapted to the circumstances. Nor does it always have to be used; the authority should decide on its use after a detailed case analysis.

The use of responsive regulation or other forms of OHS monitoring will not solve all the problems of providing workers with appropriate working conditions. In addition to the necessary changes in the law, it is vital to involve all stakeholders in the OHS process, from the OHS inspection authorities, the provider of the artificial intelligence system, the employer who uses the system, to the workers themselves involved in the work process. Only a comprehensive approach to the new OHS challenges posed by the development of artificial intelligence will allow an efficient and structured response to changing working conditions.

This study provides a starting point for a broader discussion on the regulation of occupational safety and health and methods of monitoring employers in the context of risks aris-

ing from the development of artificial intelligence. It is worth examining the responsive approach more extensively in the future with reference to specific AI risks. The changing reality calls for in-depth reflections on how to regulate it by law in order to protect workers' health.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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